

Strong Induction

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Review

To prove $P(n)$ for every integer $n \geq n_0$ by induction, prove:

1. the base case $P(n_0)$;
2. for every $k \geq n_0$, the implication

$$P(k) \Rightarrow P(k + 1).$$

The assumption $P(k)$ is the induction hypothesis.

We now introduce strong induction, which allows us to use all earlier cases.

Why use strong induction here?

For every integer $n \geq 2$, define

$P(n)$: n can be written as a product of primes.

Ordinary induction would assume $P(k)$ and try to prove $P(k+1)$. But a prime factorization of k gives no natural factorization of $k+1$.

To prove $P(60)$, the useful information comes from

$$60 = 6 \cdot 10,$$

so we need $P(6)$ and $P(10)$, not $P(59)$.

The next case may depend on earlier cases other than the immediately preceding case.

Strong induction

Let $P(n)$ be defined for every integer $n \geq n_0$.

To prove that every such $P(n)$ is true, prove:

1. **Base case:** $P(n_0)$ is true.
2. **Inductive step:** for every integer $k \geq n_0$, assume $P(n_0), \dots, P(k)$ are all true and prove $P(k+1)$.

$$(P(n_0) \wedge \dots \wedge P(k)) \Rightarrow P(k+1).$$

These assumptions form the **strong induction hypothesis**.

Ordinary induction assumes only $P(k)$. Strong induction lets us use any of the earlier cases from n_0 through k .

Why strong induction works

We do not assume all cases are true at the outset. Starting with the base case, we establish them one by one:

$$P(n_0) \qquad \text{base case}$$

$$P(n_0) \quad \therefore P(n_0 + 1)$$

$$P(n_0), P(n_0 + 1) \quad \therefore P(n_0 + 2).$$

Each step uses facts already established in earlier steps.

Define $Q(k)$ to mean that all cases through k are true:

$$Q(k) = P(n_0) \wedge \cdots \wedge P(k).$$

Base case: $Q(n_0) = P(n_0)$ is true.

Inductive step: assume $Q(k)$. The strong step gives $P(k + 1)$; together they give $Q(k + 1)$. Thus

$$Q(k) \Rightarrow Q(k + 1).$$

This is ordinary mathematical induction on $Q(k)$. We are using the principle we already know, not a new one.

Base case and induction hypothesis

Recall the proposition

$P(n)$: n can be written as a product of primes.

Base case. $P(2)$ is true because 2 is prime. A prime itself counts as a product with one prime factor.

Let $k \geq 2$. For the **strong induction hypothesis**, assume

$P(j)$ is true for every integer j with $2 \leq j \leq k$.

We must prove $P(k + 1)$.

Prime or composite

Consider the integer $k + 1$.

If $k + 1$ is prime, then it is already a product containing one prime factor. Thus $P(k + 1)$ is true.

If $k + 1$ is composite, then there are integers a and b such that

$$k + 1 = ab, \quad 2 \leq a, b < k + 1.$$

Since a and b are integers smaller than $k + 1$,

$$2 \leq a, b \leq k.$$

Therefore the strong induction hypothesis applies to both a and b .

Completing the proof

By the strong induction hypothesis, write

$$a = p_1 p_2 \cdots p_r, \quad b = q_1 q_2 \cdots q_s,$$

where every p_i and q_i is prime.

Hence

$$k + 1 = ab = p_1 p_2 \cdots p_r q_1 q_2 \cdots q_s,$$

which is a product of primes. Thus $P(k + 1)$ is true in the composite case as well.

Therefore the strong inductive step is complete. By strong induction, every integer $n \geq 2$ can be written as a product of primes.

When $k + 1$ is composite, its factors may be much smaller than k . This is why strong induction is natural here.

Summary

- ▶ In ordinary induction, the next case is proved from the immediately preceding case.
- ▶ In strong induction, the next case may use any earlier case.
- ▶ Strong induction is natural when a case reduces to much smaller cases.
- ▶ Strong and ordinary induction are logically equivalent principles.